

Sahasra Ranjan

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RESEARCH INTERESTS

Foundation models, interpretability, and robust generalization. Why do large models work, where do they fail, and how do they retain knowledge? My work spans NLP (code-switching, multilingual models), computational biology (protein-molecule interactions), and, currently, real-time speech-to-speech models: serving them under hard latency budgets and building the harnesses that expose how they fail.

EDUCATION

Indian Institute of Technology Bombay

Mumbai, India

B.Tech in Computer Science and Engineering | GPA: **9.05**/10 | Honors GPA: **9.75**/10

Aug 2019 - May 2023

HONORS & AWARDS

- **O-1A Visa**, extraordinary ability in the sciences, granted by U.S. Citizenship & Immigration Services 2026
- **ACL 2023 Outstanding Paper Award**, co-first author; awarded to less than 1% of accepted papers 2023
- **Research Excellence Award**, best Bachelor's Thesis among graduating CS students, IIT Bombay 2023
- **JEE Advanced AIR 127** / 1.2M candidates; KVPY Fellowship; NTSE Scholar 2017-2019

PUBLICATIONS

331 total citations, *h*-index 5, *i10*-index 4 (Google Scholar, Sept 2026). *denotes equal contribution.

1. Richeek Das*, **Sahasra Ranjan***, Shreya Pathak, Preethi Jyothi. *Improving Pretraining Techniques for Code-Switched NLP*. **ACL 2023** Outstanding Paper Award
Proposed boundary-aware masking strategies (Switch-MLM, Freq-MLM) that improved multilingual model performance on code-switched text by 5-7 F1 points across Hindi-English, Spanish-English, Tamil-English, and Malayalam-English benchmarks.
2. Alexander Kroll, **Sahasra Ranjan**, Martin K. M. Engqvist, Martin J. Lercher. *A general model to predict small molecule substrates of enzymes based on machine and deep learning*. **Nature Communications, 2023** (248 citations)
Developed ESP, the first general enzyme-substrate prediction model combining protein language models with molecular fingerprints; deployed as public web server (esp.cs.hhu.de) used by researchers in drug discovery and metabolic engineering.
3. Alexander Kroll, **Sahasra Ranjan**, Martin J. Lercher. *A multimodal Transformer network for protein-small molecule interactions enhances predictions of kinase inhibition and enzyme-substrate relationships* (ProSmith). **PLoS Computational Biology, 2024** (41 citations)

RESEARCH EXPERIENCE

ProSmith: Drug-Target Interaction via Multimodal Transformers

Summer 2023

Research Collaboration | Prof. Martin J. Lercher, Alexander Kroll

HHU Düsseldorf, Germany

- Extended enzyme-substrate work to drug-target interaction; designed ProSmith architecture combining bidirectional cross-attention between protein and molecule representations with gated fusion layers.
- Introduced contrastive pre-training on 500k BindingDB protein-ligand pairs, improving generalization to unseen drug scaffolds and out-of-distribution targets.
- Achieved **94.2% accuracy** on held-out test set (30% error reduction over prior SOTA), outperforming existing methods on kinase inhibition and enzyme-substrate benchmarks; published in **PLoS Comp. Bio.** (2024) and deployed at deepmolecules.org, serving **50k+ predictions**.

Gradient-based Optimization for Continual Learning

Jan 2023 - May 2023

B.Tech Thesis | Advisors: Prof. Preethi Jyothi, Prof. Manoj Prabhakaran

IIT Bombay

- Developed **CommitteeSGD**: keeps a buffer of prior examples and modifies current gradients to penalize updates that conflict with learned knowledge, reducing catastrophic forgetting. Exploited the local elasticity of neural networks (ICLR'20) to minimize decision-boundary perturbation on new examples while preserving classification of old ones.
- Achieved **3.5% accuracy improvement** on CIFAR-100 (VGG16) over standard SGD; periodic committee refresh outperformed static buffer (**66.8%** vs 63.4% test accuracy).

Improving Pretraining for Code-Switched NLP

May 2022 - Dec 2022

B.Tech Thesis | Advisor: Prof. Preethi Jyothi, CSALT Lab

IIT Bombay

- Investigated why multilingual models (mBERT, XLM-R) underperform on code-switching; identified that standard MLM ignores language boundaries, causing models to learn spurious monolingual correlations.
- Designed **Switch-MLM**: boundary-aware masking that preferentially masks tokens at language switch-points, forcing models to learn cross-lingual dependencies rather than relying on monolingual context.

- Introduced **Freq-MLM** for low-resource settings without language-ID labels; achieved **5-7 F1 improvement** on NER and sentiment analysis across Hindi-English and Spanish-English benchmarks. Published at **ACL 2023, Outstanding Paper Award** (top 1%).

Deep Learning for Enzyme-Substrate Prediction

Research Internship / Prof. Martin J. Lercher, Alexander Kroll

Summer 2021

HHU Düsseldorf, Germany

- Built one of the first multimodal transformer for enzyme-substrate prediction, jointly encoding protein sequences via ESM-1b embeddings and molecular structures via ECFP fingerprints with learned attention.
- Replaced frozen ESM-1b with end-to-end fine-tuning; improved accuracy from **82% to 91%** by allowing protein representations to adapt dynamically to substrate binding context.
- Curated 70k enzyme-substrate pairs from BRENDA, handling severe class imbalance via negative sampling from biochemically plausible non-substrates.
- Published in **Nature Communications** (248 citations); deployed at esp.cs.hhu.de, where researchers predict enzyme-substrate relationships for metabolic engineering and drug discovery.

ENTREPRENEURSHIP

Mathkraft

Co-founder, AI Math Tutor for K-8

2024 - 2025

San Francisco, CA

- Co-founded an AI tutoring startup to democratize math education; built a platform where kids create adventures and characters with math woven into the storyline. Piloted with **150+ students** at **95%+ retention** over 6 months.
- Managed 4 engineers and 6 tutors; ran 50+ hands-on sessions and found younger students need emotional connection before cognitive engagement. The human-in-loop model proved unscalable; some gaps between models and humans are structural, not accidental, which motivated my interest in alignment and robustness.

INDUSTRY EXPERIENCE

Socratic AI Inc

Founding AI Engineer (2nd engineer)

Aug 2025 - Present

Sunnyvale, CA

- **Real-time speech-to-speech inference:** built the provider layer putting Ultravox, Gemini Live, OpenAI Realtime and PersonaPlex behind one streaming protocol, so a speech model swaps in without touching the product; own turn-taking and end-of-turn VAD, barge-in, and a silence monitor separating a quiet caller from an abandoned line.
- **Latency and context under a real-time budget:** per-user tool tiering (small native direct tier + meta-tool long tail) keeps tool-call latency flat as the toolkit grows; lifted Gemini Live's 15-min audio cap via context compression and session rollover; added parallel Deepgram/Google STT decoupled from the S2S model.
- **Evaluation harness for voice agents:** skills/arms/LLM-judge/prompt-injection harness scoring agents on the prompt production actually composes, plus a tool-calling and latency sweep across realtime models, turning "the agent felt slow" into a reproducible measurement over live and recorded calls.
- **Engineering lead under the founder** on the deployment for a publicly traded enterprise communications provider, one of two multi-year, multi-million-dollar contracts the company signed (the other with a Fortune 150 semiconductor firm).
- Shipped the Cosmo Realtime SDKs (Python, TypeScript, Swift): lifecycle hooks, `useRealtimeSession()`, credential preflight. Earlier: Claude Agent SDK integration (subagents, streaming, compaction) and an Odoo ERP connector cutting 6-week enterprise migrations to days. **1,300+ commits** in the first year.

Rubrik Inc

Cloud Data Software Engineer, Data Life Cycle Team

Summer 2022 (Intern), 2023 - 2024 (Full-time)

Bengaluru, India

- Designed garbage collection for snapshot metadata; built reference tracking, mark-sweep lifecycle, and gRPC APIs for automated orphaned entry detection and removal across distributed storage nodes.
- Extended retention lock and legal hold to cloud-native workloads (archival/replication constraints for immutable backups over petabytes); implemented async SLA assignment with an event-driven poller for non-blocking parallel scheduling across thousands of protected objects.

TEACHING

Teaching Assistant, CS725: Foundations of Machine Learning (Prof. Preethi Jyothi)

Fall 2022

Teaching Assistant, CS753: Automatic Speech Recognition (Prof. Preethi Jyothi)

Spring 2023

TECHNICAL SKILLS

Languages: Python, C++, Go, TypeScript, SQL **ML/DL:** PyTorch, TensorFlow, HF Transformers, JAX, vLLM

Infra: Docker, Kubernetes, GCP, PostgreSQL

Realtime/Speech: LiveKit, WebRTC, SIP, Ultravox, Gemini Live, OpenAI Realtime, Deepgram **LLM:** Claude API / Agent SDK, RAG, LLM-as-judge harnesses